

Kindergarten Ontario Curriculum Mathematical Learning

Bigger, Smaller, Same!

Unit Overview: Bigger, Smaller, Same!

Grade: Kindergarten | **Strand:** Mathematical Learning

Target Expectations: A10.1, A10.2

Learning Goal: Children will compare objects by length, mass, and capacity using concrete materials, and describe different ways objects can be measured.

Lesson	Lesson Title	Learning Goal
Day 1	Longer and Shorter	I can compare how long things are.
Day 2	Heavier and Lighter	I can compare how heavy things are.
Day 3	Holds More, Holds Less	I can compare how much things hold.
Day 4	How Many Cubes Long?	I can measure with cubes, hands, or footsteps.
Day 5	Measurement Carnival Capstone	I can compare and measure things in different ways.

Lesson 1: Longer and Shorter

Goal: We compare LENGTH by lining things up. Longer, shorter, or same length.

Expectation: A10.2

Learning Goal: "I can compare how long things are."

Materials: M1_length_strips, M3_object_set, M7_measure_chart, M9_animal_set

1. Minds On (7 mins): Max Compares Trails

Max: Look at this trail of footprints. Is it LONGER or SHORTER than this other trail?

Bring children to the carpet. Max + Tess puppets. 'I'm Max the Measure Moose. Today we learn LONGER, SHORTER, and SAME.'

2. Action (20 mins): Length Stations

- Whole group: compare 4 pairs of objects.
- Three stations.
- Stretch challenge at Station C: find 3 things the same length as your hand.
- Discuss the line-up rule.
- Circulate to record diagnostic evidence on the length tracker.

3. Consolidation (8 mins): Max's Length Brag

Discuss:

- What does LONGER mean?
- What does SHORTER mean?
- How do you compare two things?
- When are two things the SAME length?

Exit: Refrain chant 3x.

Lesson 2: Heavier and Lighter

Goal: We compare MASS by feeling weight or using a balance. Heavier, lighter, or same weight.

Expectation: A10.2

Learning Goal: "I can compare how heavy things are."

Materials: M2_paper_balance, M3_object_set, M7_measure_chart, M10_measurement_poster

1. Minds On (7 mins): Tess Tests Heavy and Light

Tess: I am LIGHT. Max is HEAVY. Look — when I sit on this seesaw, Max goes DOWN.

Bring children to the carpet. Max + Tess puppets. 'Yesterday we compared LENGTH. Today we compare MASS — HEAVY or LIGHT.'

2. Action (20 mins): Mass Stations

- Whole group: hand-feel and balance 4 pairs of objects.
- Three stations.
- Stretch challenge at Station B: order 4 objects from heaviest to lightest using the balance.
- Discuss when hand-feel works and when balance is needed.
- Circulate to record formative evidence on the mass tracker.

3. Consolidation (8 mins): Tess's Mass Brag

Discuss:

- What does HEAVIER mean?
- What does LIGHTER mean?
- When do you use a balance?
- How does the balance show heavier?

Exit: Refrain chant 3x.

Lesson 3: Holds More, Holds Less

Goal: We compare CAPACITY by pouring. Holds more, holds less, or same.

Expectation: A10.2

Learning Goal: "I can compare how much things hold."

Materials: M3_object_set, M4_cup_cards, M7_measure_chart, M10_measurement_poster

1. Minds On (7 mins): Max Pours Between Cups

Max: This big cup is full of beans. Will all the beans fit in this small cup? Let's see!

Bring children to the carpet. Max + Tess puppets. 'Yesterday we compared MASS. Today we compare CAPACITY — HOW MUCH it HOLDS.'

2. Action (20 mins): Capacity Stations

- Whole group: pour and compare 4 pairs of containers.
- Three stations.
- Stretch challenge at Station B: order 5 cups from smallest to largest capacity by pouring.
- Discuss the pour rule.
- Circulate to record formative evidence on the capacity tracker.

3. Consolidation (8 mins): Max's Capacity Brag

Discuss:

- What does HOLDS MORE mean?
- What does HOLDS LESS mean?
- How do you compare two cups?
- Tall and thin vs short and wide — which always wins?

Exit: Refrain chant 3x.

Lesson 4: How Many Cubes Long?

Goal: We measure with non-standard UNITS — cubes, hands, footsteps.
Same object: many units.

Expectation: A10.1

Learning Goal: "I can measure with cubes, hands, or footsteps."

Materials: M1_length_strips, M5_linking_cubes, M7_measure_chart, M8_unit_chart

1. Minds On (7 mins): Max Measures with Cubes

Max: My measuring stick says 5 CUBES LONG. Can YOUR pencil be 5 cubes long?

Bring children to the carpet. Max + Tess puppets. 'For 3 days we COMPARED. Today we MEASURE — count UNITS.'

2. Action (20 mins): Measurement Stations

- Whole group: measure 4 objects with cubes.
- Three stations.
- Stretch challenge at Station A: measure the same object twice, once with cubes, once with hand-spans.
- Discuss unit choice.
- Circulate to record formative evidence on the measurement tracker.

3. Consolidation (8 mins): Max's Measurement Brag

Discuss:

- What does MEASURE mean?
- Why must cubes TOUCH?
- When is hand-spans better than cubes?
- When is footsteps better than hand-spans?

Exit: Refrain chant 3x.

Lesson 5: Measurement Carnival Capstone

Goal: We use length, mass, and capacity to compare and measure. We pick the right way for each job.

Expectation: A10.1, A10.2

Learning Goal: "I can compare and measure things in different ways."

Materials: M1_length_strips, M2_paper_balance, M3_object_set, M4_cup_cards

1. Minds On (7 mins): Max and Tess at the Carnival

Max: Welcome to the MEASUREMENT CARNIVAL! Three stations: LENGTH, MASS, CAPACITY. Pick your favourite!

Bring children to the carpet. Max + Tess in carnival hats. 'CAPSTONE day! We use everything: LENGTH, MASS, CAPACITY, MEASURING.'

2. Action (20 mins): The Measurement Carnival

- Hand out carnival templates.
- Station A (LENGTH): Order 5 ribbons from shortest to longest.
- Station B (MASS): Use balance to find heaviest of 5 objects.
- Station C (CAPACITY): Pour to find which holds most of 4 cups.
- Circulate to record summative evidence on the capstone rubric.

3. Consolidation (8 mins): Max's Carnival Brag

Discuss:

- Which station was your favourite — and why?
- How do you compare length?
- How do you compare mass?
- How do you compare capacity?
- Which way is best for measuring a desk?

Exit: Refrain chant 3x.

Name: _____

Date: _____

Max's Length Hunt

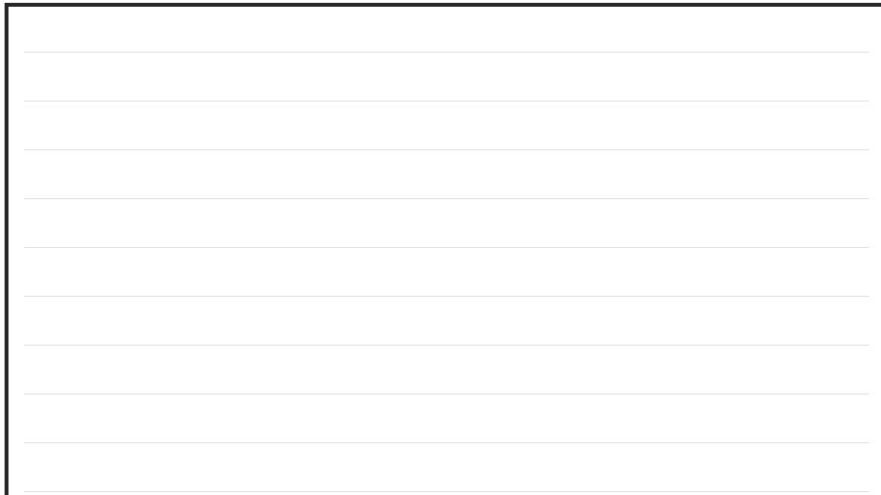
Learning Goal: I can compare how long things are.

Mission

1. Look at each pair.
2. Number the strips from 1 (shortest) to 4 (longest).
3. Look around your classroom.

Part 3: Find SAME-length things!

Look around your classroom. Find 2 things that are the SAME length. Draw them.

A large rectangular box with a black border, containing ten horizontal lines for drawing. The lines are evenly spaced and extend across the width of the box.

Name: _____

Date: _____

Tess's Mass Match

Learning Goal: I can compare how heavy things are.

Mission

1. Look at each pair.
2. Look at each balance.
3. Look around your classroom.

Part 3: Find heavier in your classroom!

Look around your classroom. Pick 2 things. Draw them. Circle the HEAVIER one.

Name: _____

Date: _____

Max's Capacity Pour

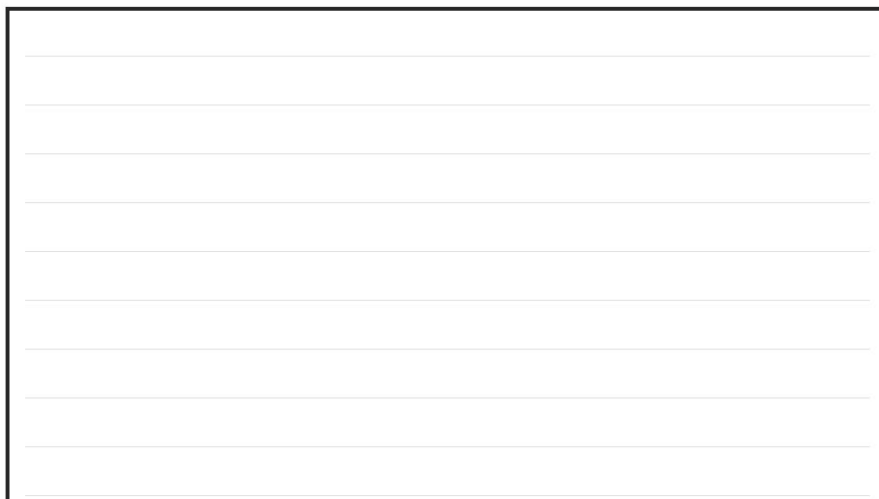
Learning Goal: I can compare how much things hold.

Mission

1. Look at each pair.
2. Number the cups from 1 (holds least) to 4 (holds most).
3. Think about your kitchen at home.

Part 3: Find more in your kitchen!

Think about your kitchen at home. Pick 2 things that hold liquid. Draw them. Circle the one that HOLDS MORE.

A large rectangular box with a black border, containing ten horizontal lines for drawing. The lines are evenly spaced and extend across the width of the box.

Name: _____

Date: _____

Max's Cube Counting

Learning Goal: I can measure with cubes, hands, or footsteps.

Instructions for the Student:

1. Each picture shows an object with cubes lined up beside it.
2. Look at each object.
3. Pick 3 things at home.

Part 1: Count the cubes!

Each picture shows an object with cubes lined up beside it.
Count the cubes. Write the number.

Name: _____

Date: _____

Max's Carnival Capstone

Learning Goal: I can compare and measure things in different ways.

Instructions for the Student:

1. Number the strips 1 (shortest) to 4 (longest).
2. Look at each balance.
3. Circle the cup that holds MORE in each pair.

Part 3: CAPACITY carnival!

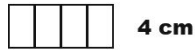
Circle the cup that holds MORE in each pair. Then find a SAME-capacity pair and circle BOTH cups.

Length Comparison Strips

(M1_length_strips)

A set of 24 paper strips of varied lengths used by pairs to sort, compare, and order by length.

Centimetre Strips



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 2

Laminate: yes

USE IN CLASS

Quantity: 10 sets

Lessons: 1, 4, 5

Prep: ~25 min

TEACHER PREP

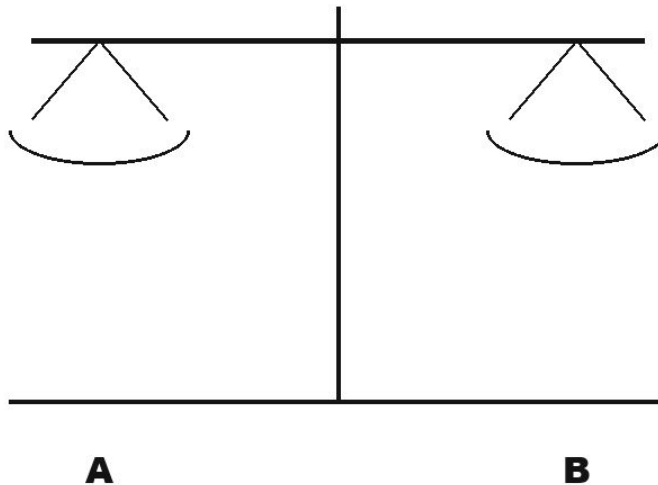
1. Print 10 sets total (1 set per pair).
2. Laminate before cutting for repeated use.
3. Cut along the horizontal solid lines into 24 individual strips per set.
4. Store each set in a labelled zip-top bag.

Paper Balance Mat

(M2_paper_balance)

A letter-size mat showing a balance with two pans for visualizing mass comparison.

Balance Scale



PRINT SPECS

Size: letter landscape
Colour: bw
Pages per set: 1
Laminate: yes

USE IN CLASS

Quantity: 10-12 mats
Lessons: 2, 5
Prep: ~15 min

TEACHER PREP

1. Print 12 mats (1 per pair plus 2 extras).
2. Laminate every mat for dry-erase use.
3. Distribute one mat per pair; keep extras at the teacher table.

Comparison Object Cards

(M3_object_set)

A set of 24 paper object tokens (heavy/light, big/small) for sorting activities across length, mass, and capacity comparisons.

Comparison Object Cards	
1	Book _____
2	Pencil _____
3	Brick _____
4	Feather _____

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 2

Laminate: yes

USE IN CLASS

Quantity: 10 sets

Lessons: 1, 2, 3, 5

Prep: ~25 min

TEACHER PREP

1. Print 10 sets (1 per pair).

2. Laminate before cutting.

3. Cut along the grid lines into 24 individual tokens per set.

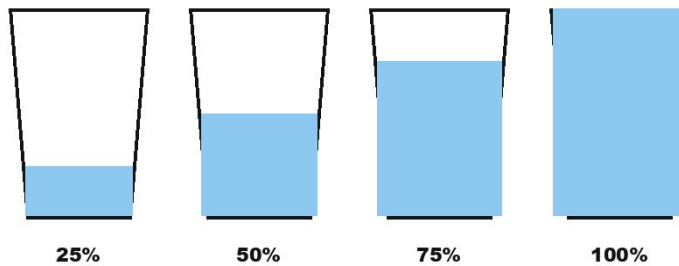
4. Store each set in a labelled zip-top bag.

Capacity Cup Cards

(M4_cup_cards)

A set of 12 paper cup cards in 3 sizes (small, medium, large) for capacity sorting activities.

Cups & Capacity



PRINT SPECS

Size: letter portrait
Colour: bw
Pages per set: 1
Laminate: yes

USE IN CLASS

Quantity: 10 sets
Lessons: 3, 5
Prep: ~20 min

TEACHER PREP

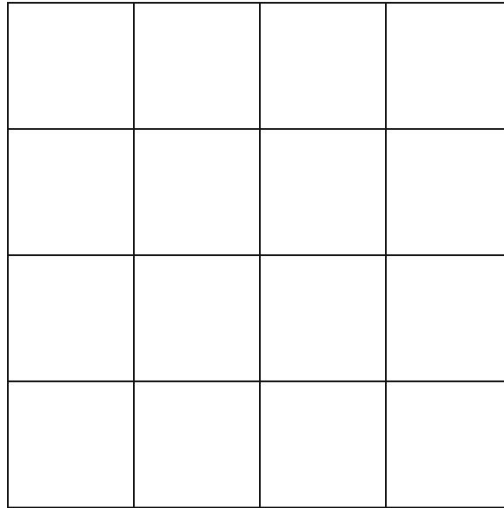
1. Print 10 sets (1 per pair).
2. Laminate before cutting.
3. Cut along the grid lines into 12 individual cups per set.
4. Store each set in a labelled zip-top bag.

Paper Linking Cubes

(M5_linking_cubes)

A set of 60 paper cubes that link to measure length in non-standard units.

Area Grid — 4 × 4



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 2

Laminate: yes

USE IN CLASS

Quantity: 10 sets

Lessons: 4, 5

Prep: ~30 min

TEACHER PREP

1. Print 10 sets (1 per pair).

2. Laminate before cutting for durability.

3. Cut along the grid lines into 60 individual cubes per set.

4. Store each set in a labelled zip-top bag.

Carnival Capstone Template

(M6_carnival_template)

A letter-size template for the Lesson 5 Measurement Carnival capstone with 3 station response areas (length, mass, capacity).

Carnival Capstone Template	
1	Length _____
2	Mass _____
3	Capacity _____
4	Thin Section Dividers _____

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: no

USE IN CLASS

Quantity: 25 templates

Lessons: 5

Prep: ~5 min

TEACHER PREP

1. Print 25 templates (1 per child plus 5 extras).

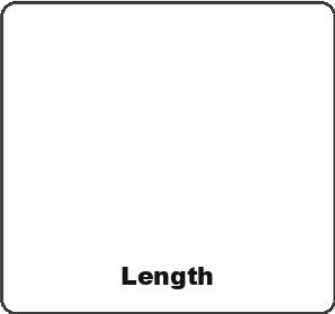
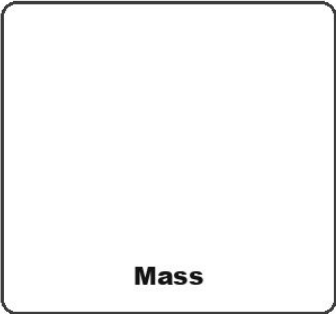
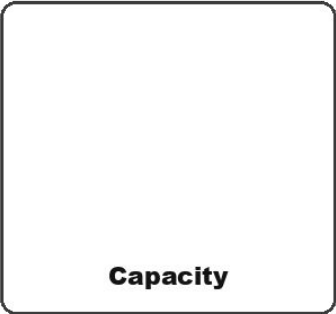
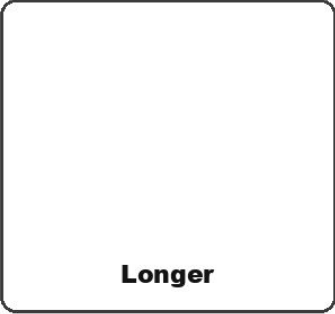
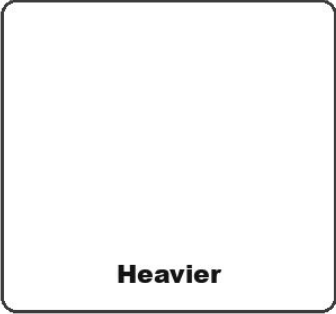
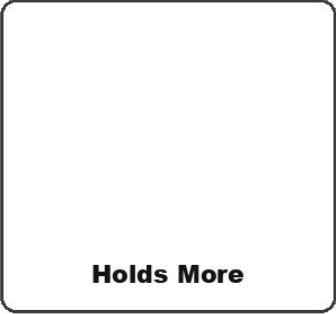
2. Stack near the teacher table for distribution at the start of Lesson 5 Action.

3. No lamination needed — these are single-use summative artifacts.

Measurement Anchor Chart

(M7_measure_chart)

A teacher-display anchor chart showing length, mass, and capacity comparisons with examples.

Measurement Anchor Chart		
 Length	 Mass	 Capacity
 Longer	 Heavier	 Holds More

PRINT SPECS

Size: letter landscape

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 chart

Lessons: 1, 2, 3, 4

Prep: ~10 min

TEACHER PREP

1. Print 1 copy on standard letter paper (or scale to 11x17 if available).

2. Laminate for repeated reference across the unit.

3. Tape to the front classroom wall before Lesson 1.

Non-Standard Units Chart

(M8_unit_chart)

A teacher-display anchor chart of non-standard units (cubes, hand-spans, footsteps) with example uses.

Non-Standard Units Chart	
1	Cubes _____
2	Hand-Spans _____
3	Footsteps _____
4	A Hand-Span _____

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 chart

Lessons: 4, 5

Prep: ~10 min

TEACHER PREP

1. Print 1 copy on standard letter paper.

2. Laminate for repeated reference.

3. Tape next to the Measurement Anchor Chart on the front wall before Lesson 4.

Comparison Animal Tokens

(M9_animal_set)

A set of 12 paper animal tokens (mouse, cat, dog, bear, elephant) of graded sizes for length comparison.

Comparison Animal Tokens	
1	Tokens grouped by species _____
2	Mice _____
3	Cats _____
4	Elephants _____

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 10 sets

Lessons: 1, 5

Prep: ~20 min

TEACHER PREP

1. Print 10 sets (1 per pair).

2. Laminate before cutting.

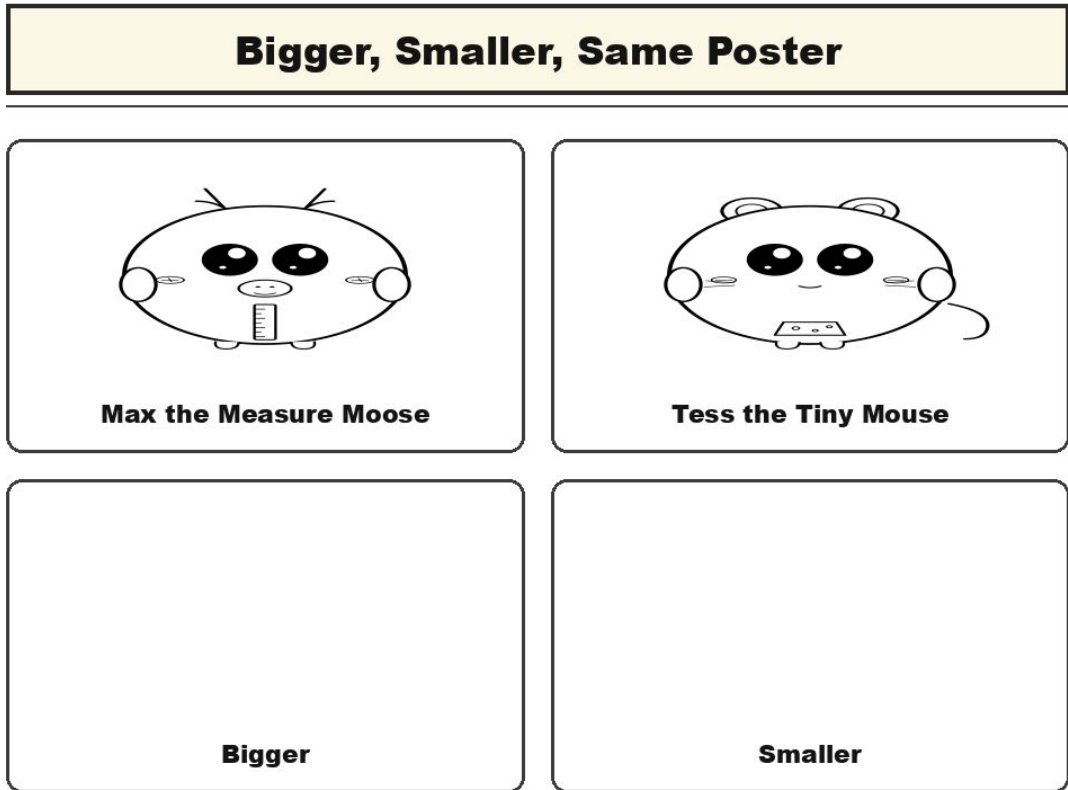
3. Cut into 12 individual tokens per set.

4. Store each set in a labelled zip-top bag.

Bigger, Smaller, Same Poster

(M10_measurement_poster)

A unit anchor poster featuring Max and Tess with examples of all three measurement attributes.



PRINT SPECS

Size: letter landscape

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 poster

Lessons: 1, 2, 3, 4, 5

Prep: ~10 min

TEACHER PREP

1. Print 1 copy on letter or 11x17 paper.

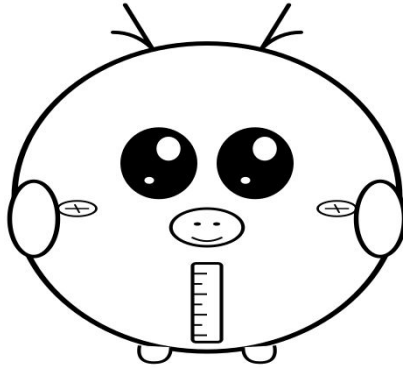
2. Laminate the poster for the full unit duration.

3. Tape to the front classroom wall on Day 1 before Lesson 1 begins.

Max Puppet

(char_max_puppet)

A paper puppet of Max the Measure Moose used by the teacher in the Minds-On phase of every lesson to introduce concepts in child voice.



Max the Measure Moose

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 puppet

Lessons: 1, 2, 3, 4, 5

Prep: ~12 min

TEACHER PREP

1. Print 1 copy of Max on cardstock if possible.

2. Cut around the outside of Max's silhouette including the bottom tab.

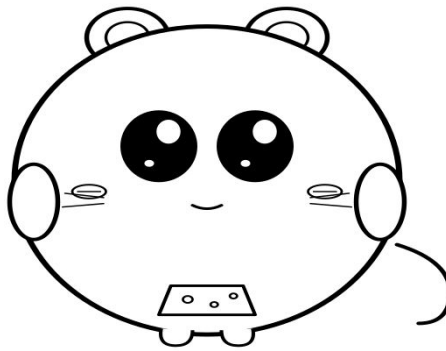
3. Laminate before assembly for durability.

4. Glue or tape a popsicle stick behind the bottom tab so Max can be held up.

Tess Puppet

(char_tess_puppet)

A paper puppet of Tess the Tiny Mouse used by the teacher in all 5 lessons to embody the 'small' side of comparisons.



Tess the Tiny Mouse

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 puppet

Lessons: 1, 2, 3, 4, 5

Prep: ~12 min

TEACHER PREP

1. Print 1 copy of Tess on cardstock if possible.

2. Cut around the outside of Tess's silhouette including the bottom tab.

3. Laminate before assembly for durability.

4. Glue or tape a popsicle stick behind the bottom tab so Tess can be held up.

Name: _____

Date: _____

Summative Assessment Rubric

Expectation	Level 1 (Beginning)	Level 2 (Developing)	Level 3 (Achieving)	Level 4 (Exceeding)
A10.1 describe different ways that objects can be measured	Cannot yet describe a way to measure. Says 'I don't know' or 'big' without a unit reference.	Names one way to measure (cubes) when prompted. Counts cubes with adult help.	Independently names cubes, hand-spans, and footsteps as ways to measure. Picks an appropriate unit for an object's size.	Names multiple non-standard units AND can describe length using two different units to verify (e.g., '5 cubes OR 1 hand-span').
A10.2 compare the length, area, mass, and capacity of objects using concrete materials	Compares only by sight; struggles with one of the three attributes (length, mass, or capacity). Cannot use a tool (balance, pour) consistently.	Compares length and one other attribute with prompts. Uses the balance or pours with adult support.	Independently compares all three attributes (length, mass, capacity) using appropriate techniques: line up for length, balance for mass, pour for capacity. Recognizes 'same' as a valid result.	Orders 4+ objects across all three attributes AND verifies using a second technique (e.g., compares two cups by both pouring AND counting cubes that fit).

Junior Measure Captain Certificate

This certificate is proudly presented to:

For successfully completing the 5-Day Bigger Smaller
Same unit and demonstrating the skills of a Measure
Captain. A measurement certificate. Max the Measure
Moose congratulates the student.

By completing this unit, this student has demonstrated:

- Comparing lengths by lining up
- Comparing mass with hand-feel and balance
- Comparing capacity by pouring
- Measuring with cubes, hands, and footsteps

Marketplace Listing

A 5-day Kindergarten unit on measurement comparison framed around Max the Measure Moose and Tess the Tiny Mouse.

Price: \$8.99 CAD

Pages: ~42

Classroom time: 175 min

Teacher prep: ~200 min

Top tags: Kindergarten, math, measurement, length, mass, capacity, comparing, non-standard units

What's included:

- Unit blueprint with curriculum mapping (1 file)
- 5 lesson plans with minds-on, action, consolidation, expanded walkthroughs (5 files)
 - 5 student worksheets with answer keys and image placeholders (5 files)
 - Manipulatives plan with 12 print-ready assets (1 file: length strips, paper balance, object cards, cup cards, linking cubes, carnival template, anchor charts,...)
 - Formative + reflection sheets (1 file)
 - Assessment suite (1 file: diagnostic tracker, 3 formative trackers, summative rubric, summative administration guide, Junior Measure Captain certificate)